

FW Self-Level: Horizon Mode — Implemented

Status: Implemented (option A) — both MR and FW. **Date:** 2026-08-05 **Scope:** UAVXArmQ attitude control (control.c, control.h, rc.c, rc.h).

1. Decision

Reinstate **Horizon Mode** as a 3rd AttitudeMode (between Angle and Rate), applied to **both** multicopter and fixed-wing. Rejected the alternative of augmenting RateMode with a new self-level parameter — functionally identical feel, but required a new gain param and duplicated legacy logic.

A[Roll].R.Kp (tag 0) / A[Pitch].R.Kp (tag 5) are unchanged rate-P gains. No parameter was renamed or added. pHorizonTransScale (tag 30) is reused as-is and is already visible on the GCS.

2. Changes

control.h:51

```
enum AttitudeModes { AngleMode, HorizonMode, RateMode, UnknownAttitudeMode };
```

rc.c:1203 — **mode selection**

Widened the Ch5/pot mapping * 1.5f → * 3.0f so the Aux1 pot sweeps Angle → Horizon → Rate (Horizon = middle). CurrMaxRollPitchStick made non-static (:276), extern added in rc.h:82.

control.c — **MR (DoQuaternionAttitudeControl)**

AttitudeMode == RateMode unchanged. Otherwise compute the quaternion angle error Qa[] once, then branch: - **AngleMode:** existing angle PI (2·Qa·P.Kp + IntE). - **HorizonMode:** P.IntE = 0, then c AngleRateMix = Limit(1.f - CurrMaxRollPitchStick * pHorizonTransScale, 0.f, 1.f); A[a].R.Desired = Limit1(2.f*Qa[a]*A[a].P.Kp, A[a].R.Max) * AngleRateMix + Threshold(A[a].Stick, pStickDeadZone) * A[a].R.Max * (1.f - AngleRateMix); Yaw keeps its existing stick/hold handling.

control.c — **FW (DoFWAttitudeControl, refactor of the old inline switch)**

Same structure as MR for Pitch/Roll: Angle / Horizon / Rate cases. The old inline switch (AttitudeMode) in DoControl() was replaced by a single DoFWAttitudeControl(dT) call (:665).

3. Dead code removed (quaternion/ROC-era)

- ControllingAltitudeROC() + #if PIC_ROC_CTRL gate + the #define — the abandoned **MR ROC-based altitude hold** (was laggy; replaced by the position-PI cascade).
- ConditionIntE() — defined but never called.

4. Retained

- `DoROCControl()` — still live for the **FW BoostClimb** path (`control.c:256`): rapid full-RC climb then motor-off glide between WPs (soaring cross-country).
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5. Build note

`params.c` `FLOAT/U8` macros used a parameter named `cls`, which collided with the `ParamMetaEntry` member `.cls` — the preprocessor substituted the argument into the member name, producing `.classExplicit = ....` Fixed by renaming the macro parameters to `pcls` (`params.c:124-127`).